

The Economics of Optional Royalties:
Platform Competition, Enforcement, and Routing
in NFT Markets
Final Presentation

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Theoretical Model: Environment & Agents

- ▶ **Market Timeline:** A three-period discrete time framework.
 - ▶ $t = 0$: Platforms set trading fees (β_m), royalty regimes, and enforcement credibility (κ_m).
 - ▶ $t = 1$: Primary market sale occurs; the Creator sets primary price (p_I), royalty rate (α), and mint location (d).
 - ▶ $t = 2$: Secondary market opens; traders make routing and royalty-payment decisions.
- ▶ **Agents:** A single Creator (C) and a continuum of traders (total measure of 1).
- ▶ **Trader Heterogeneity:** Traders differ on two independent dimensions.
 - ▶ *Information*: Informed (anticipate exact future value) vs. Uninformed (use noisy signals).
 - ▶ *Motive*: Speculators (profit-driven) vs. Collectors (value long-term consumption and creator support).

Theoretical Model: Platform Policies & Routing

- ▶ **Platform Regimes:** Marketplaces choose a royalty policy $R_m \in \{\text{mandatory, optional, zero}\}$.
- ▶ **Enforcement Credibility:** Platforms choose parameter $\kappa_m \in [0, 1]$, representing the expected fraction of royalties actually collected, incurring a convex cost $C_m(\kappa_m)$.

- ▶ **Effective Transaction Costs:** For a trade on platform m , the effective proportional charge is:

$$\tau_{m,t}(p) = \beta_m + d_{tm}\kappa_m\alpha$$

where $d_{tm} \in \{0, 1\}$ indicates if the trader pays the royalty.

- ▶ **Switching Costs:** Moving a trade away from the minting platform d incurs a price-dependent friction cost $k(p_{II}) \geq 0$.
- ▶ **Routing Mechanism:** Traders allocate volume across platforms via a logit choice model based on expected utility and a routing elasticity parameter (λ_t) .

Proposition 1: Monopoly Platform

- ▶ **Setup:** A single marketplace with mandatory royalties ($\alpha > 0$) serving both Speculators and Collectors, isolating trader heterogeneity from competitive routing.
- ▶ **The Trade-off:** Increasing enforcement credibility (κ) makes the platform attractive to Collectors, but raises costs and deters Speculators while incurring direct enforcement costs.
- ▶ **Result:**
 - ▶ If the marginal revenue gain from enforcing royalties at $\kappa = 0$ exceeds the marginal enforcement cost $C'(0)$, an optimal interior enforcement choice exists where $\kappa^* \in (0, 1]$.
 - ▶ If the valuation of enforcement by Collectors is too weak, the platform optimally chooses $\kappa^* = 0$.

Proposition 2: The Race to the Bottom

- ▶ **Setup:** A symmetric duopoly featuring mandatory royalties, identical fees, only Speculator traders, and zero switching costs.
- ▶ **Mechanism:** Speculators are purely profit-driven and route entirely based on total transaction costs.
- ▶ **Dynamics:** Any positive difference in enforcement causes the platform with lower enforcement to capture the entire market volume discontinuously.
- ▶ **Result:** The unique Nash equilibrium is for both marketplaces to set enforcement credibility to zero ($\kappa_A^* = \kappa_B^* = 0$).
- ▶ **Implication:** Without switching frictions or collector-driven demand, competitive platform pressure forces the complete elimination of promised royalty payments.

Proposition 3: Collectors Counterbalance Competition

- ▶ **Setup:** A symmetric duopoly with mandatory royalties, identical fees, but now trading is populated by both Speculators and Collectors.
- ▶ **Mechanism:** Because Collectors value community and creator support, they are less sensitive to pure cost differences, creating upward pressure on enforcement that counteracts the profit motives of Speculators.
- ▶ **Result 1 (Symmetric):** A symmetric interior equilibrium ($\kappa_A^* = \kappa_B^* = \kappa^* \in (0, 1)$) exists if the marginal revenue gained from retaining Collectors precisely offsets the marginal revenue lost from Speculators and the marginal cost of enforcement.
- ▶ **Result 2 (Asymmetric):** If responses are sufficiently non-monotone, an asymmetric equilibrium ($\kappa_A^* \neq \kappa_B^*$) may emerge where one platform caters to Collectors (high enforcement) and the other to Speculators (low enforcement).

Proposition 4: The Impact of Switching Costs

- ▶ **Home-Bias in Routing:** Positive switching costs ($k(p_{II}) > 0$) generate strict home-platform bias even when platforms are otherwise identical. At symmetric enforcement ($\kappa_d = \kappa_n$):

$$s_{d,S} > \frac{1}{2}, \quad s_{d,K} > \frac{1}{2}$$

- ▶ **Speculative Undercutting Weakened:** Stronger own enforcement reduces speculative routing, but switching costs make this penalty less severe:

$$\frac{\partial s_{d,S}}{\partial \kappa_d} < 0, \quad \frac{\partial^2 s_{d,S}}{\partial \kappa_d \partial k} > 0 \quad \text{at } \kappa_d = \kappa_n$$

- ▶ **Collector-Side Threshold:** Collector routing increases in own enforcement if and only if collectors value enforcement more than the effective royalty cost:

$$\phi_{\kappa} > \alpha p_{II}$$

Proposition 4: Volume & Equilibrium Effects

- ▶ **Secondary Volume Effect:** Routed secondary volume increases in own enforcement whenever the collector-side gain dominates the speculative-routing loss.
- ▶ **Comparative Statics for Equilibrium:** If switching costs raise the marginal payoff to enforcement ($\Pi_{\kappa k} > 0$), then:

$$\frac{d\kappa^*(k)}{dk} > 0$$

- ▶ **Implication:** Positive switching costs increase the range of parameters under which non-zero enforcement can be sustained. They weaken speculative undercutting while preserving collector and creator-side gains.

Fundamental Empirical Measures

- **Collection-Level Royalty Rate (α_c):** Anchored to the 95th percentile of realized rates on OpenSea for collection c :

$$\alpha_c = \text{Percentile}_{0.95} \left(\frac{\text{Royalty Fee}_i}{\text{Price}_i} \right) \text{ s.t. Platform} = \text{OpenSea}$$

- **Enforcement Credibility ($\kappa_{m,c,t}$):** The realized fraction of expected royalties collected by platform m in week t :

$$\kappa_{m,c,t} = \frac{\sum \text{Royalty Fee Paid}_{m,c,t}}{(\alpha_c/100) \cdot \sum \text{Secondary Volume}_{m,c,t}}$$

- **Effective Total Cost Rate ($\tau_{m,c,t}$):** The proportional sum of platform fees (β) and enforced royalties:

$$\tau_{m,c,t} = \beta_{m,c,t} + (\alpha_c \cdot \kappa_{m,c,t})$$

Market Share and Switching Proxies

- ▶ **Platform Market Share ($s_{m,c,t}$):** Calculated as the volume share within a specific collection-week:

$$s_{m,c,t} = \frac{\text{Volume}_{m,c,t}}{\sum_{j \in \{Blur, OpenSea\}} \text{Volume}_{j,c,t}}$$

- ▶ **Mint Platform Proxy (d_c):** The platform where the earliest secondary trade is observed in the sample:

$$d_c = \operatorname{argmin}_{\text{platform}} (\text{Block Time}_i) \text{ for collection } c$$

- ▶ **Collection-Week Switching Share ($\text{Switch_Share}_{c,t}$):** The proportion of total volume executed off the identified minting platform:

$$\text{Switch_Share}_{c,t} = \frac{\sum_{m \neq d_c} \text{Volume}_{m,c,t}}{\text{Total Volume}_{c,t}}$$

- ▶ **Mint Indicator:** Binary flag $I(m = d_c)$ used to identify whether platform m is the collection's origin for calculating switching costs $k(p_{II})$.

Panel Dataset Structure

Sample Observations: One or the other Platform Data (Week of 2023-04-03) in ETH

Week	Platform	Share (Vol)	α_c (%)	$\kappa_{m,c,t}$	$\tau_{m,c,t}$ (%)
2023-04-03	Blur	0.515	9.50	0.947	10.77
2023-04-03	OpenSea	0.485	9.50	0.320	3.68
2023-04-03	Blur	0.973	5.00	0.594	3.21
2023-04-03	OpenSea	0.027	5.00	0.450	2.98
2023-04-03	Blur	0.173	6.90	0.597	5.21

Variable Definitions:

- ▶ **Market Share ($s_{m,c,t}$):** Proportion of collection volume on platform m . This is used as the target moment for routing behavior.
- ▶ **Enforcement Credibility (κ):** Realized vs. stated royalties ($\frac{\text{Paid}}{\alpha \cdot \text{Vol}}$). This captures the optionality of the regime.
- ▶ **Effective Cost (τ):** Total proportional friction ($\beta + \alpha\kappa$) controlling trader routing elasticity.

MSM Estimator: Model Setup

Effective trading costs

$$\tau_{Bm} = \beta_{Bm} + \alpha_m \kappa_{Bm}, \quad \tau_{Om} = \beta_{Om} + \alpha_m \kappa_{Om}$$

Standardized log prices

$$z_{Bm} = \frac{\log p_{Bm} - \overline{\log p_B}}{\sigma(\log p_B)}, \quad z_{Om} = \frac{\log p_{Om} - \overline{\log p_O}}{\sigma(\log p_O)}$$

Off-mint switching costs

$$k_B(m) = k_0 + k_1 z_{Bm}, \quad k_O(m) = k_0 + k_1 z_{Om}$$

Switching-adjusted effective costs

$$\tilde{\tau}_{Bm} = \tau_{Bm} + \mathbf{1}\{B \text{ not mint}\} k_B(m), \quad \tilde{\tau}_{Om} = \tau_{Om} + \mathbf{1}\{B \text{ mint}\} k_O(m)$$

$$\Delta \tilde{\tau}_m = \tilde{\tau}_{Bm} - \tilde{\tau}_{Om}, \quad \Delta \kappa_m = \kappa_{Bm} - \kappa_{Om}$$

Type-specific utility indices

$$V_{Sm} = \delta_B - \lambda \Delta \tilde{\tau}_m$$

$$V_{Cm} = \delta_B - \lambda \Delta \tilde{\tau}_m + \phi_K \Delta \kappa_m$$

MSM Estimator: Choice Probabilities and Objective

Logit choice probabilities

$$s_{Bm}^S = \frac{1}{1 + e^{-V_{Sm}}}, \quad s_{Bm}^C = \frac{1}{1 + e^{-V_{Cm}}}$$

Mixture share

$$\hat{s}_{Bm} = \gamma_S s_{Bm}^S + (1 - \gamma_S) s_{Bm}^C, \quad \hat{s}_{Om} = 1 - \hat{s}_{Bm}$$

Model outcomes

$$\widehat{\text{share gap}}_m = \hat{s}_{Bm} - \hat{s}_{Om} = 2\hat{s}_{Bm} - 1$$

$$\widehat{\text{switch}}_m = \begin{cases} 1 - \hat{s}_{Bm}, & \text{if Blur is mint} \\ \hat{s}_{Bm}, & \text{if Blur is not mint} \end{cases}$$

MSM objective

$$\hat{\theta} = \arg \min_{\theta} r(\theta)' r(\theta)$$

where

$$r(\theta) = M^{\text{model}}(\theta) - M^{\text{data}}.$$

Moments Matched in the MSM Estimator

The estimator chooses parameters to match a set of empirical moments computed from the data and from the model.

1. Binned moments

- ▶ Mean share gap by royalty bin:
- ▶ Mean share gap by enforcement-difference bin:
- ▶ Mean share gap by effective-cost-difference bin:
- ▶ Mean switching share by effective-cost-difference bin:

2. Mean and subgroup moments

- ▶ Overall mean Blur share and mean switching share
- ▶ Means split by mint vs. non-mint
- ▶ Means split by high-price vs. low-price

3. Slope moments

- ▶ Slope of share gap against $\tau_B - \tau_O$
- ▶ Slope of switching against $\tau_B - \tau_O$
- ▶ Slope of share gap against $\kappa_B - \kappa_O$

Which Moments Matter Most?

(1) Share-gap moments by effective cost difference

$$E[s_B - s_O \mid \tau_B - \tau_O \in \text{bin}]$$

These are crucial for identifying cost sensitivity λ .

(2) Switching moments by effective cost difference

$$E[\text{switch} \mid \tau_B - \tau_O \in \text{bin}]$$

These are the key moments for the switching-cost parameters k_0 and k_1 .

(3) Share-gap moments by enforcement difference

$$E[s_B - s_O \mid \kappa_B - \kappa_O \in \text{bin}]$$

and the slope:

$$\text{slope}(s_B - s_O, \kappa_B - \kappa_O)$$

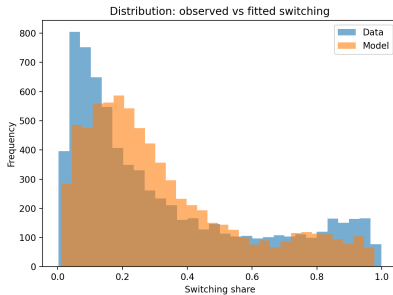
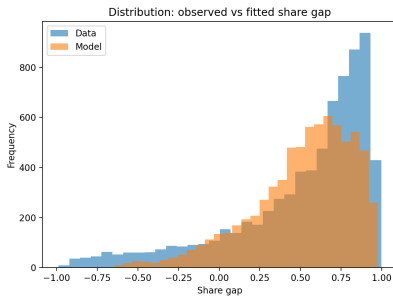
These are the most important moments for the enforcement-preference parameter ϕ_K .

Bootstrap Confidence Intervals

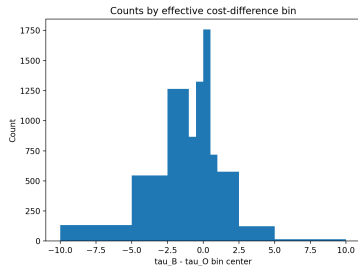
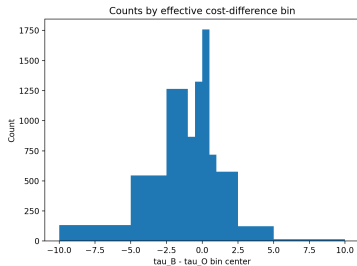
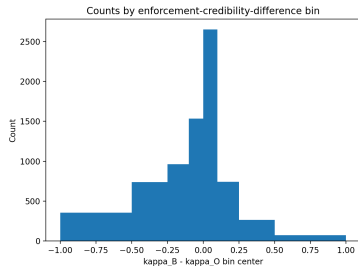
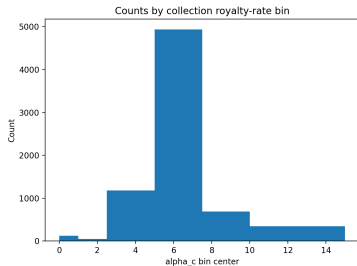
Parameter	Baseline	Boot Mean	Boot SD	2.5%	50%	97.5%
δ_B	0.917	0.922	0.032	0.866	0.924	0.977
λ	0.199	0.182	0.046	0.081	0.177	0.285
ϕ_K	3.382	3.452	2.325	-1.885	3.659	7.522
k_0	1.605	1.707	0.701	1.151	1.638	3.166
k_1	5.000	4.813	0.436	3.480	5.000	5.000
γ_S	0.790	0.783	0.169	0.198	0.827	0.928

Notes: Table reports baseline MSM estimates and bootstrap summary statistics based on 200 successful bootstrap replications. Confidence intervals are percentile intervals. Analyzed on 7,400 markets.

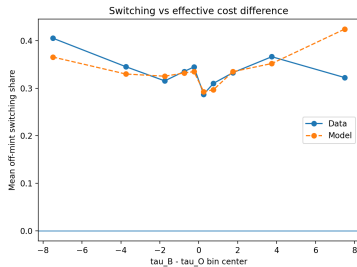
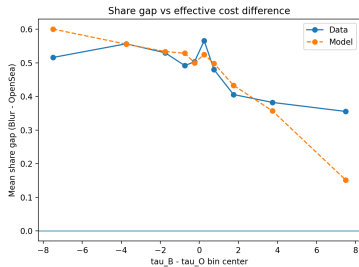
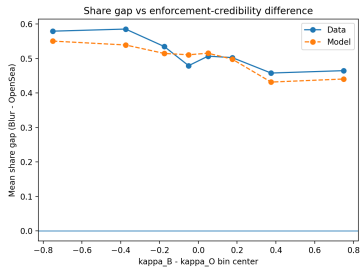
Distributional Results



General Characteristics



Model Fit



Counterfactual Results: Top-Line Effects

Scenario	Δ Mean s_B	Δ Mean Share Gap	Δ Mean Switch
Equalize enforcement	0.0009	0.0018	-0.0023
OpenSea matches Blur enforcement	0.0032	0.0065	-0.0024
Full enforcement both	0.0009	0.0018	-0.0023
Remove switching frictions	-0.0599	-0.1199	0.0815
Blur matches OpenSea enforcement	-0.0023	-0.0047	0.0001
Zero enforcement preference	0.0075	0.0151	-0.0034
Zero price-dependent switching	-0.0219	-0.0439	0.0219
Targeted full enforcement, high royalty	-0.0005	-0.0011	-0.0002

Notes: Entries report average changes relative to the baseline model prediction.

Additional Counterfactuals: Enforcement Taste and Targeted Enforcement

Scenario	Market Structure	<i>n</i>	Δ Mean s_B
Zero enforcement preference	Medium, Blur stronger, Blur mint	819	-0.0226
Zero enforcement preference	Medium, OpenSea stronger, Blur nonmint	306	0.0509
Zero enforcement preference	High, Blur stronger, Blur mint	112	-0.0280
Zero enforcement preference	High, OpenSea stronger, Blur nonmint	64	0.0448
Targeted full enforcement (high royalty only)	High, Blur stronger, Blur mint	112	0.0449
Targeted full enforcement (high royalty only)	High, Blur stronger, Blur nonmint	15	0.0514
Targeted full enforcement (high royalty only)	High, OpenSea stronger, Blur mint	221	-0.0304
Targeted full enforcement (high royalty only)	High, OpenSea stronger, Blur nonmint	64	-0.0514

Interpretation: Eliminating the direct collector taste for enforcement mainly matters in high-royalty markets. Targeting full enforcement only to high-royalty markets reproduces nearly all of the meaningful enforcement effects.

Additional Counterfactuals: Price-Dependent Switching

Scenario	Market Structure	n	Δ Mean s_B
Zero price-dependent switching	High price, Blur mint, non-parity	1805	-0.0875
Zero price-dependent switching	High price, Blur mint, tau parity	1505	-0.0935
Zero price-dependent switching	High price, Blur nonmint, non-parity	200	0.1410
Zero price-dependent switching	High price, Blur nonmint, tau parity	152	0.1501
Remove switching frictions	High price, Blur mint, non-parity	1805	-0.1448
Remove switching frictions	High price, Blur mint, tau parity	1505	-0.1538
Remove switching frictions	High price, Blur nonmint, non-parity	200	0.2041
Remove switching frictions	High price, Blur nonmint, tau parity	152	0.2191

Interpretation: The price-dependent switching term is very important. Turning off the price slope creates large changes in high-price markets, but obviously less than eliminating all switching frictions.

Takeaways from the Additional Counterfactuals

- ▶ **The enforcement-taste channel matters.** Setting $\phi_K = 0$ moves Blur share in exactly the cells where enforcement should matter most:
 - ▶ Blur loses share when Blur initially has the enforcement advantage
 - ▶ Blur gains share when OpenSea initially has the enforcement advantage
- ▶ **The economically relevant enforcement effects are concentrated in high-royalty markets.** Targeted full enforcement only in high-royalty markets produces almost all of the meaningful enforcement effects.
- ▶ **Switching is very important in high-price markets.** Removing only the price-dependent part of switching costs function causes large reallocations of market share, especially in high-price mint and non-mint markets.

What comes next for me?

I will be starting a Pre-Doc this summer at the Becker Friedman Institute for Economics at UChicago working with Professor Alexander Torgovitsky on topics in Microeconometrics, applied econometrics, and causal inference.

Questions?